ROBOTICS SIMULATION INTERN / TECHNICAL REPORT

Dual-Rail Robotic Car Painting Cell

Two powered UR5e rail manipulators with an independent overhead XYZ/tilt gantry | Webots R2025a

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1. Overview and cell architecture

This project simulates coordinated exterior painting of an approximately 4.8 × 1.9 × 1.6 m simplified vehicle. Two six-axis UR5e arms on independent, physically motorized prismatic rails paint opposite side panels concurrently. A third, Cartesian XYZ slider gantry with a servo-tilted nozzle approaches roof, bonnet, boot lid and rear tailgate surfaces. Tool positions are sensor checked before blue virtual coating is displayed; no paint-fluid dynamics are modeled.

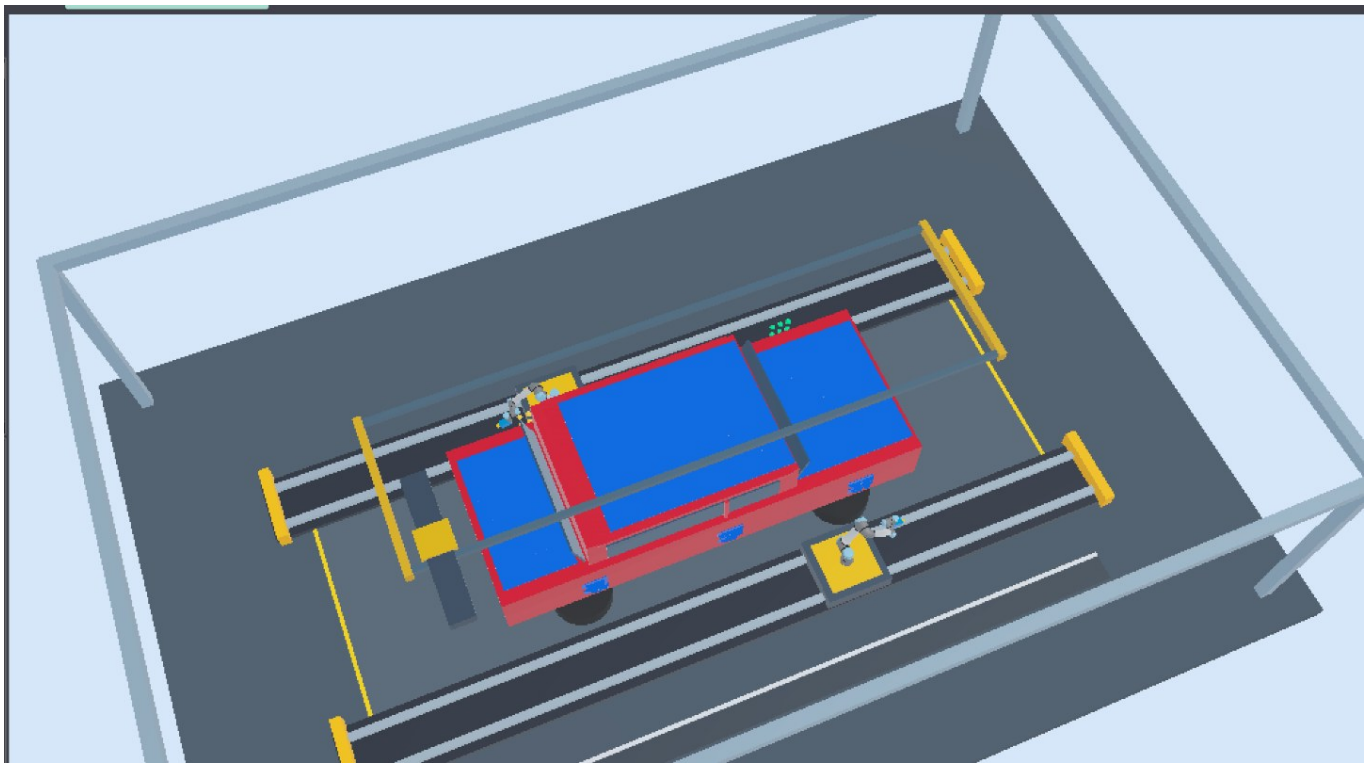


Figure 1. Webots Stage 23 full-world overhead view showing both rail robots and visible blue coating on upper surfaces. Screenshot supplied from candidate-run simulation.

Component	Implementation
Simulation	Webots R2025a; Python robot controllers; 32 ms basic time step
Side manipulators	Two UR5e six-joint articulated arms; independently actuated SliderJoint rails
Overhead	XYZ gantry with three LinearMotors and a tilt RotationalMotor (not a third UR5e)
Coating	Quality-gated blue visual geometry; virtual spray / modeled dose proxies, not measured film thickness

2. Motion planning and closed-loop control

Side-surface raster planning

Each side is divided into three stations along vehicle length. At each stop, the arm traverses a 3 × 5 serpentine raster patch (15 end-effector targets). Thus each side has 45 commanded painting poses in the full run. Two independent rail controllers communicate readiness, arrival and completion to their respective arms. The rail is held stationary during arm tracking, and encoder and carriage-GPS readings are used to verify station position.

UR5e inverse kinematics and pose verification

The six joint motors use an iterative damped-least-squares Jacobian controller that regulates measured nozzle-tip XYZ and spray-axis direction. Two GPS devices estimate the nozzle axis. Desired tool directions point toward the appropriate side surface (+Y near side, -Y far side). A nominal standoff near 0.25 m is targeted. Coating is credited only when the pose, axis and standoff gates pass; calibration and homing motion do not generate paint.

Overhead surface planner

The overhead controller uses deterministic, retract–translate–tilt–approach motion. It retracts the spray head before lateral travel, then lowers or tilts it toward the selected surface. The controller verifies motor encoders, tip GPS and nominal surface gap before emitting an individually identified blue visual paint tile. Rear-tailgate poses use a 90° nozzle tilt and a rear-facing approach.

Surface	Full-mode target grid	Qualified in supplied log
Roof	6 rows × 8 columns = 48	48 / 48
Bonnet/hood	6 rows × 6 columns = 36	36 / 36
Boot lid	6 rows × 4 columns = 24	24 / 24
Rear tailgate	4 rows × 5 columns = 20	Not shown in supplied log

Supervision and basic collision awareness

- Independent rail state machines enforce homing, settle checks, station handshakes and motion hold during painting.
- Tool-body axis-aligned box and conservative robot-link centerline-proxy clearance checks operate during side-arm motion.
- A modeled proxy-clearance stop threshold of 0.100 m and warning threshold of 0.140 m are applied; these are not full mesh-clearance or safety-certified assessments.
- Fault handling stops painting and sends a shared cell-stop radio message; Stage 16 tested an injected sensor reading, not a real collision.

3. Results from candidate Webots runs

The following metrics are transcribed from the candidate-provided Stage 23 console and screenshots. They describe simulated trajectories and visual coverage, not paint thickness or production throughput.

Subsystem / metric	Evidence	Status
Near-side UR5e	3/3 rail stations; 45/45 Cartesian raster poses	PASS
Far-side UR5e	3/3 rail stations; 45/45 Cartesian raster poses	PASS
Side paint visualisation	Blue coating events and patch coverage logging	Observed
Overhead homing	GANTRY_HOME_VERIFIED	PASS
Roof	48/48 qualified targets	PASS
Bonnet/hood	36/36 qualified targets	PASS
Boot lid	24/24 qualified targets	PASS
Tailgate	20 planned targets; no final result in supplied console	Unverified
Experimental WBT parser	Duplicate dampingConstant field in original 4 Stage 23 worlds	Corrected in accompanying revised package

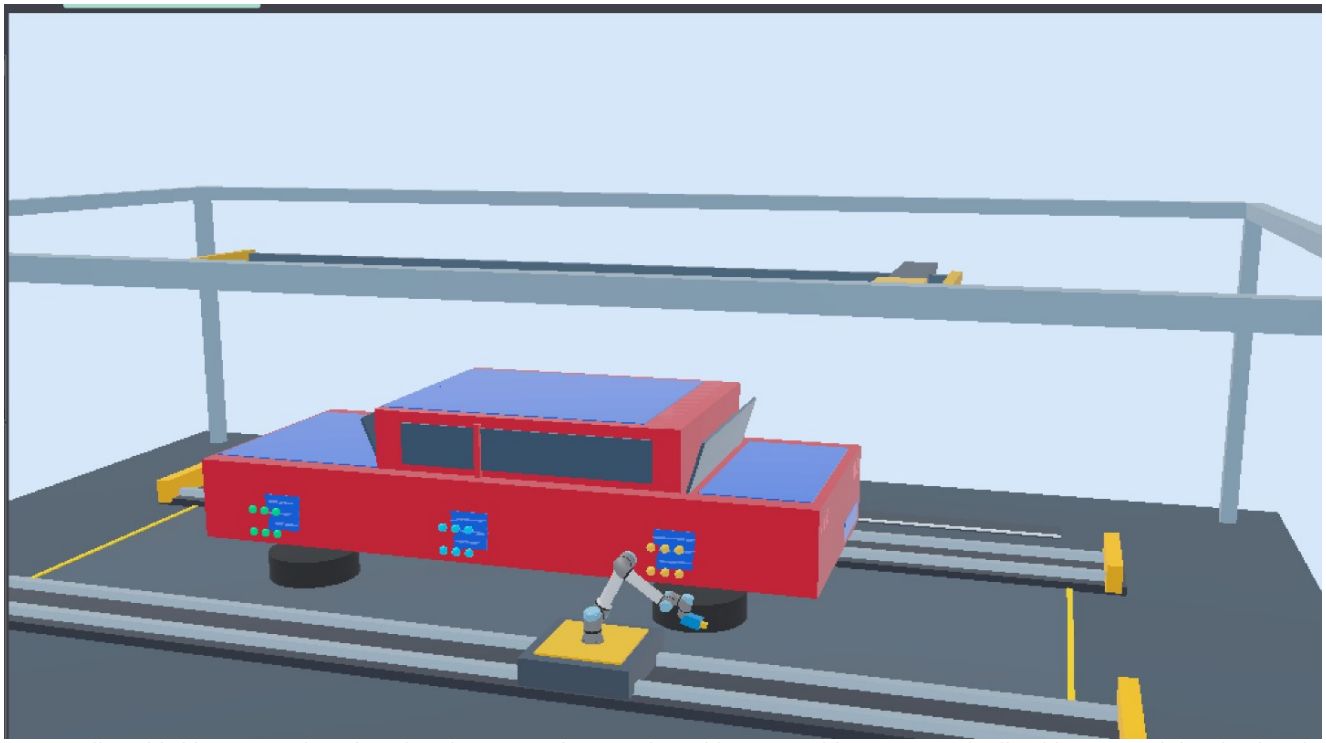


Figure 2. Full-world side perspective after virtual upper-surface coating. Side-arm positions, powered rail guideways and the simulated overhead hardware are visible.

Offline checks: 33 Python tests passed in the revised package. A rate-limited actuator mock completed all 30 quick-mode overhead targets. These checks do not replace a clean Webots R2025a run. The original video log also contains an H.264 two-pass bitrate convergence error despite a subsequent “creation finished” message.

4. Limitations, reproducibility and conclusions

Interpretation and limitations

- The vehicle is a simplified static box-based model, not a production car CAD mesh with verified gaps, doors, edges or paintable trim masks.
- Blue patches are virtual display geometry. The dose map uses an assumed relative-dose model; there is no droplet transport, overspray, material flow, calibration or physical film-thickness validation.
- Side rails model motorized movement. The default overhead variant uses motor-actuated kinematic joints without moving-part Physics nodes; the experimental dynamics variant is not independently Webots validated.
- Centerline proxy clearances are conservative screening measurements, not proof of full articulated-link, booth or vehicle collision freedom; industrial operation would require independent engineered safeguards.
- Dual-rail concurrency is demonstrated. A same-workload, independently measured single-rail baseline is not yet established; do not describe an estimated 2× ratio as demonstrated industrial cycle-time improvement.

Reproduce the simulation

Software: Webots R2025a and Python 3.10. Extract Simulation_Files and open worlds/CarPaintingCell_Stage23_AllSurfaces_Quick.wbt, reset and run. For full raster sampling open CarPaintingCell_Stage23_AllSurfaces_Full.wbt. Check “GANTRY_HOME_VERIFIED”, all four “PANEL … qualified” summaries, both side-rail “FINISHED” messages and no safety abort. The experimental dynamics worlds are optional and should not be the primary evaluated demonstration.

Evidence required to finalize the full run

The provided complete console excerpt ends after “PANEL boot : qualified 24 / 24”. A full Stage 23 success claim requires a later “PANEL tailgate : qualified 20 / 20” and “FINISHED: roof=48/48 hood=36/36 boot=24/24 tailgate=20/20” message. It also requires a replayable 3–7 minute demo video and the project, screenshots and final report deposited in the submission folder.

Conclusion

The candidate demonstrates practical sensor-feedback motion control, multi-robot coordination, powered side-axis motion, joint-proxy safety screening, deterministic raster planning and quality-gated multi-surface coating visualisation. The observed side-arm and three upper-panel results are successful in simulation. Tailgate completion, clean parser startup and final video playback remain explicit submission-gate checks.

Evidence basis: Stage 23 console logs supplied on 23 October 2026, three Webots screenshots, source controllers and offline tests included with this accompanying project. This is a technical report draft pending final tailgate verification.